



Machine learning-based prediction model for omental metastasis in right-sided colon cancer patients: a retrospective multicenter study

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Abstract

Purpose Current diagnostic modalities lack sufficient sensitivity for detecting omental metastasis (OM), often underestimating metastatic burden. Unlike traditional statistical model, machine learning (ML) model is designed to detect subtle variable interactions and model nonlinear patterns that traditional statistics overlook, enhancing the reliability of OM risk evaluation in clinical practice. The aim of the study was to build a ML model in preoperatively predicting OM in right-sided colon cancer (RCC) patients using a multicenter dataset.

Methods This retrospective multicenter study included 1798 RCC patients: 1206 from Zhejiang Cancer Hospital (training set $n=804$, test set $n=402$) and 592 from the Second Affiliated Hospital of Harbin Medical University (validation set). OM status, tumor location, preoperative CEA level, preoperative CA199 level, Grade, histology, tumor size and age of patients were recorded. Six ML models including extreme gradient boosting (XGB), artificial neural network (ANN), logistic regression (LR), random forest (RF), support vector machine (SVM) and decision tree (DT) were developed for the OM prediction in RCC. The area under the receiver operator characteristic (ROC) curve (AUC), accuracy, sensitivity, specificity, precision, F1 score and decision curve analysis (DCA) were analyzed for judging predictive performance.

Results The OM rates in training set, test set and validation set were 10.4%, 9.5% and 10.0%, respectively. The XGB model outperforming five other algorithms (ANN, RF, LR, SVM, and DT) across training set (AUC=0.924, 0.096 gain vs LR), internal test (AUC=0.868, 0.038 gain vs LR) and validation set (AUC=0.766, 0.065 gain vs LR). The comparison of accuracy, sensitivity, specificity, precision and F1 score revealed the XGB model exhibited the best performance. The DCA curve also suggested that XGB had better clinical decision-making capability than the other five models. Feature importance analysis highlighted preoperative CEA level and tumor location as key predictors.

Conclusion Our study developed and validated an XGB-based machine learning model that could accurately predict OM in RCC patients using routine preoperative variables. This model demonstrates strong discriminative ability and clinical utility, assisting personalized risk stratification and appropriate treatment decisions.

Keywords Omental metastasis · Right-sided colon cancer · Machine learning · Risk model

Introduction

In colon cancer, omental metastasis (OM) is a crucial indicator, characterized by significantly poorer prognosis and distinct therapeutic implications compared to non-metastatic disease [1–3]. Patients with OM typically exhibit

reduced long-term survival rates, with aggressive disease behavior and high recurrence rates [1, 4]. Moreover, significantly different from patients without OM, the management for patients with OM often require more extensive therapeutic intervention such as cytoreductive surgery and hyperthermic intraperitoneal chemotherapy [1, 5]. Currently, imaging examinations exhibit relatively low diagnostic efficacy in detecting OM. These modalities, including computed tomography (CT) and magnetic resonance imaging (MRI), often struggle to accurately identify early or subtle omental lesions due to the complex anatomical

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structure of the omentum and the indistinct boundaries between metastatic foci and surrounding adipose tissue [6], leading to underdiagnosis and potential delays in optimal treatment planning.

Conventional multivariate logistic regression (LR) is insufficient for predicting OM in cancer patients. On the one hand, it relies on the core assumption that each predictor's relationship with OM log-odds is linear, yet clinical/biological links to OM risk are rarely linear. On the other hand, it fails to handle variable interaction effects, while OM risk often stems from variable relationships. These flaws stop LR capturing defining OM risk patterns, leaving it unable to meet precise risk stratification needs, a gap machine learning models address [7–9]. The ML algorithm is capable of identifying intricate nonlinear relationships among clinical features, which are difficult to detect using conventional statistical methods [10–12]. In addition, machine learning directly complements conventional imaging's limitations in OM detection. Traditional imaging struggles with tiny OM lesion, which blend with omental adipose tissue, creating blurred boundaries that human visual interpretation misses. Machine learning algorithm could integrate preoperative variables and generate a quantifiable omental metastasis risk that serves as an independent reference to support clinical decision-making, which could reduce diagnostic biases caused by relying solely on imaging and thus effectively addresses its inaccuracies in detection.

Right-sided colon cancer (RCC) exhibits unique biological and clinical features that foster a particularly intimate relationship with omental metastasis. Unlike left-sided tumors, RCC possesses distinct molecular profiles, which contribute to their aggressive phenotype [13–15]. RCC originates from the midgut (distinct from left-sided hindgut tumors), enabling lymphatic and vascular connections to the omentum, creating pathways for tumor dissemination [16]. Besides, RCC often presents at advanced stage with larger tumors, making it more likely to penetrate the serosa and directly invade the omentum [17–19]. Hence, whether metastasis occurs in the omentum for RCC patients is a crucial point needs to be taken into consideration when making treatment decisions. However, current diagnostic modalities lack sufficient sensitivity for detecting OM, often underestimating metastatic burden. Therefore, there is a need for prediction tools with high accuracy to assist clinicians in assessing the risk of OM in RCC patients, as the current routine examination provides limited diagnostic assistance.

In the current study, we aimed to build a ML model with high predictive accuracy in preoperatively predicting OM in RCC patients using the multicenter datasets, enabling accurate risk stratification, with the potential

to standardize treatment decisions and improve patient outcomes.

Methods

Study population

This retrospective study included RCC patients at the Zhejiang Cancer Hospital (Center 1) from November 2006 to May 2024 for model training set and test set (2:1 ratio). The external validation cohort comprised patients from the the Second Affiliated Hospital of Harbin Medical University (Center 2) treated between January 2019 and December 2024. Inclusion criteria included: (1) aged ≥ 18 years; (2) patients with histologically confirmed right-sided colon cancer; (3) histological types were limited to adenocarcinoma and mucous/signet-ring cell carcinoma. Exclusion criteria included: patients with unknown OM status, tumor location, preoperative CEA level, preoperative CA199 level, Grade, histology, tumor size and age.

Variables selection

Given that the primary purpose of this predictive model was to preoperatively predict OM in RCC patients, candidate variables were strategically selected from clinically accessible preoperative indicators. These indicators were routinely collected during standard clinical practice, ensuring the model's practicality and feasibility for preoperative application. The variable selection process involved two key steps: first, univariate analysis was performed to retain variables with statistical significance ($p < 0.05$), as these were considered potential predictors of OM; second, multicollinearity among the retained variables was assessed using heatmaps to ensure independence between variables, avoiding interference with model stability.

Through above screening method, the final candidate variables included tumor location, preoperative carcinoembryonic antigen (CEA) level, preoperative carbohydrate antigen 199 (CA199) level, tumor grade, histological type, and tumor size. All patient data were de-identified prior to analysis, complying with institutional data protection regulations and HIPAA guidelines.

Development of the predictive model

Six algorithms including extreme gradient boosting (XGB), artificial neural network (ANN), logistic regression (LR), random forest (RF), support vector machine (SVM) and

decision tree (DT) were selected to strike a balance between interpretability and performance. Simpler models (LR and DT) provide high interpretability, as clinicians can trace how key variables drive omental metastasis (OM) predictions, while more complex ones (XGB and RF) deliver stronger performance by capturing nonlinear and interactive OM risk relationships. Complemented by models like ANN and SVM to cover diverse methodological approaches, this selection ensures the framework meets clinical needs for transparency while retaining robust predictive accuracy for real-world OM risk assessment.

The XGB is an optimized distributed gradient-boosting library that innovates the transformation of weak learners into strong learners, significantly increasing classification accuracy [20]. The ANN is a mathematical model that simulates neurons processing information. It has strong generalization ability and robustness and can provide a high data processing efficiency [21]. The LR model uses the maximum likelihood function method and gradient descent to solve the parameters to achieve the purpose of binary classification of data, which is a classical method for data classification [22]. The RF model is a decision tree based on a bagging frame work, which can deal with high-dimensional feature input samples and has excellent accuracy [23]. The SVM model is a super vised learning algorithm often used to solve classification problems [24]. The DT is represented by a tree structure and can mathematically predict the best choice. The DT is often used to solve problems in biological sciences due to its ease of understanding [25].

Candidate variables were input into six ML models for the OM prediction in RCC (Fig. 1). In this study, the training set was used to build ML models and cross-validation, and the test and validation sets were used to evaluate the efficiency and robustness of the prediction model.

Machine learning models were developed using a modular programming approach with well-established machine learning packages. Six classification models were

constructed: logistic regression (glm function of base (v 4.5.1)), random forest (randomForest (v 4.7.1.2)), support vector machine (e1071 (v 1.7.16)), decision tree (rpart (v 4.1.24)), artificial neural network (nnet (v 7.3.20)), and XGBoost (xgboost (v 1.7.11.1)). All models used a consistent data structure with status as the binary outcome variable, and standardized preprocessing was applied where applicable. 10 × tenfold cross-validation was used to optimize hyperparameters and control overfitting, ensuring model robustness. The best model for each ML method was fitted on the training set using these variables and optimal hyperparameters.

Evaluation of the predictive model

The area under the receiver operator characteristic (ROC) curve (AUC) was used to evaluate the models' discrimination power. Calculated the accuracy, sensitivity, specificity, precision and F1 score of several models as auxiliary indicators for judging predictive performance. The decision curve analysis (DCA) was used to assess model fit and clinical benefits. Calibration plots were used to evaluate the agreement between predicted probabilities and observed omental metastasis (OM) event rates. SHapley Additive exPlanations (SHAP) was used to interpret the model's predictions. SHapley Additive exPlanations (SHAP) was employed to interpret the model's predictions. SHAP values were computed to measure the contribution of each feature to omental metastasis (OM) risk. Based on these metrics' performance on training and test sets, the best predictive model was selected for visualization and evaluated on the external validation set (Fig. 1).

Statistical analysis

Statistical analyses were performed using R version 4.5.1. Categorical variables were one-hot encoded and

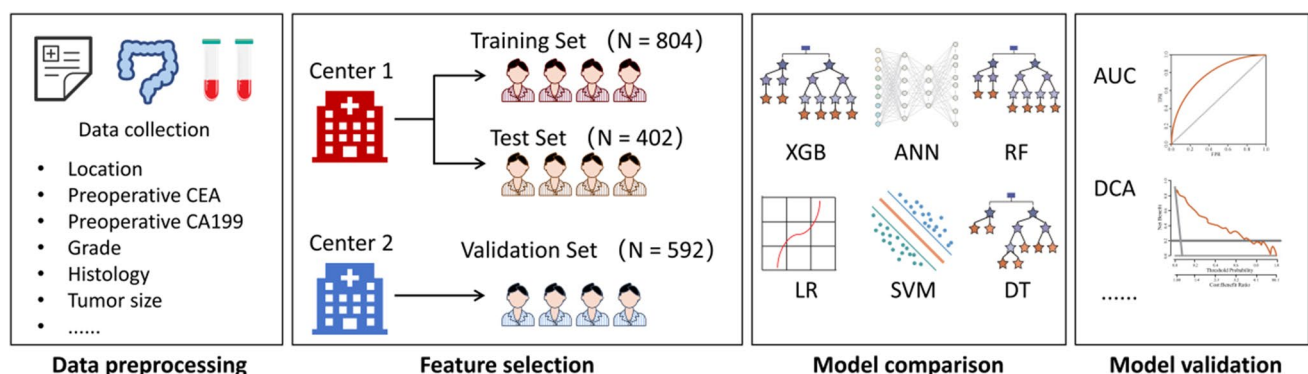


Fig. 1 Workflow of the main steps in this study

continuous variables were standardized to z-scores. Continuous variables were presented as means and standard deviation. Independent-sample t-tests were used for normally distributed continuous variables; in contrast, the Mann–Whitney U test was used for non-normally

distributed continuous variables. Categorical variables are expressed as frequencies and percentages, and comparisons were made using the chi-square test or Fisher's exact test. Statistical significance was set at $p < 0.05$.

Results

Clinical features

A total of 1798 CRC patients were enrolled in the study (1206 patients from Zhejiang Cancer Hospital and 592 patients from the Second Affiliated Hospital of Harbin Medical University). The 1206 patients were randomly divided into training set ($n = 804$) and test set ($n = 402$) with a ratio of 2:1, and the 592 patients were analyzed as validation set. The OM rates in training set, test set and validation set were 10.4%, 9.5% and 10.0%, respectively. More detailed data could be found in Table 1. Besides, we also assessed the correlation of the variables with a heatmap, and found that there was no significant correlation and no collinearity, and variables were independent of each other (Fig. 2a-c).

Feature selection

In the univariate analysis, tumor location ($P = 0.007$ for training set, $P = 0.010$ for test set and $P = 0.002$ for validation set), preoperative CEA ($P < 0.001$ for training set, $P = 0.002$ for test set and $P = 0.005$ for validation set), preoperative CA199 ($P < 0.001$ for training set, $P = 0.003$ for test set and $P = 0.022$ for validation set), grade ($P < 0.001$ for training set, $P < 0.001$ for test set and $P < 0.001$ for validation set), histology ($P < 0.001$ for training set, $P < 0.001$ for test set and $P = 0.001$ for validation set) and tumor size ($P < 0.001$ for training set, $P = 0.001$ for test set and

Table 1 Clinicopathological characteristics of the training, test and validation sets

Characteristics	Training Set ($N = 804$)	Test Set ($N = 402$)	Validation Set ($N = 592$)
Gender			
Female	387 (48.1%)	184 (45.8%)	283 (47.8%)
Male	417 (51.9%)	218 (54.2%)	309 (52.2%)
Location			
Cecum	200 (24.9%)	97 (24.1%)	160 (27.0%)
Ascending colon	304 (37.8%)	158 (39.3%)	222 (37.5%)
Hepatic flexure	117 (14.6%)	46 (11.4%)	64 (10.8%)
Transverse colon	183 (22.8%)	101 (25.1%)	146 (24.7%)
Preoperative CEA (ng/ml)			
0–5	471 (58.6%)	248 (61.7%)	347 (58.6%)
> 5	333 (41.4%)	154 (38.3%)	245 (41.4%)
Preoperative CA199 (U/ml)			
0–37	576 (71.6%)	281 (69.9%)	422 (71.3%)
> 37	228 (28.4%)	121 (30.1%)	170 (28.7%)
Grade			
I–II	553 (68.8%)	266 (66.2%)	413 (69.8%)
III–IV	251 (31.2%)	136 (33.8%)	179 (30.2%)
Histology			
Adenocarcinoma	723 (89.9%)	374 (93.0%)	530 (89.5%)
Mucinous/signet ring-cell	81 (10.1%)	28 (6.97%)	62 (10.5%)
Tumor size (cm)			
< 5	453 (56.3%)	220 (54.7%)	295 (49.8%)
≥ 5	351 (43.7%)	182 (45.3%)	297 (50.2%)
Age (years)	61.5 (12.2)	61.8 (12.5)	63.0 (11.8)
Omental metastasis	84 (10.4%)	38 (9.5%)	59 (10.0%)

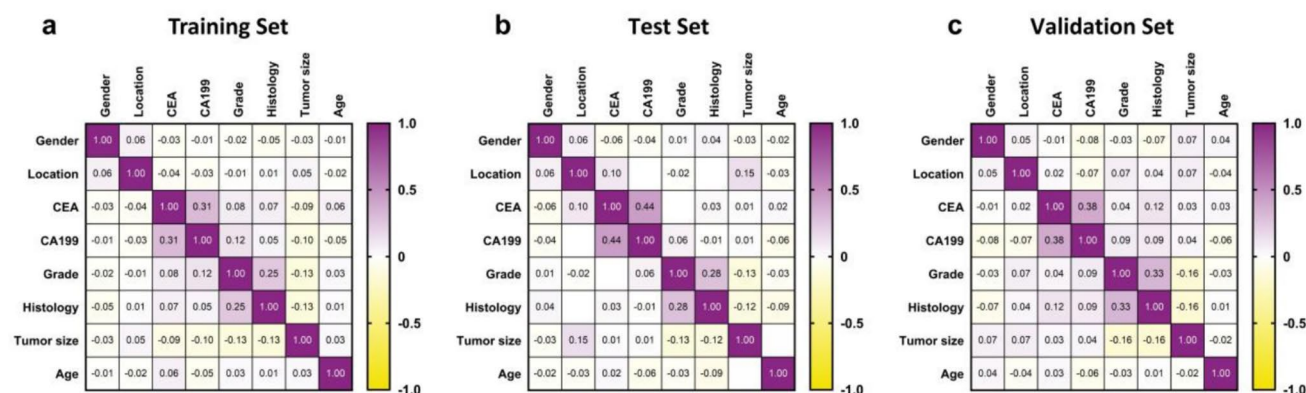
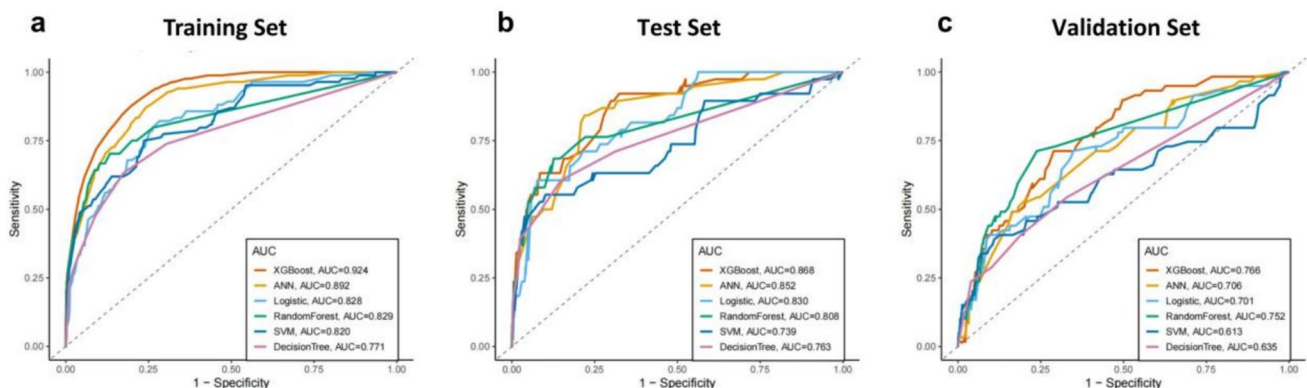


Fig. 2 Correlation analysis heatmap between feature variables in (a) Training set, (b) Test set, and (c) Validation set

Table 2 Univariate analysis for OM

Characteristics	Training Set (N = 804)			Test Set (N = 402)			Validation Set (N = 592)		
	Non-OM (N = 720)	OM (N = 84)	P	Non-OM (N = 364)	OM (N = 38)	P	Non-OM (N = 533)	OM (N = 59)	P
Gender			0.479			0.971			0.847
Female	343 (47.6%)	44 (52.4%)		166 (45.6%)	18 (47.4%)		256 (48.0%)	27 (45.8%)	
Male	377 (52.4%)	40 (47.6%)		198 (54.4%)	20 (52.6%)		277 (52.0%)	32 (54.2%)	
Location			0.007			0.010			0.002
Cecum	179 (24.9%)	21 (25.0%)		89 (24.5%)	8 (21.1%)		147 (27.6%)	13 (22.0%)	
Ascending colon	282 (39.2%)	22 (26.2%)		150 (41.2%)	8 (21.1%)		209 (39.2%)	13 (22.0%)	
Hepatic flexure	107 (14.9%)	10 (11.9%)		42 (11.5%)	4 (10.5%)		57 (10.7%)	7 (11.9%)	
Transverse colon	152 (21.1%)	31 (36.9%)		83 (22.8%)	18 (47.4%)		120 (22.5%)	26 (44.1%)	
CEA (ng/ml)			< 0.001			0.002			0.005
0–5	445 (61.8%)	26 (31.0%)		234 (64.3%)	14 (36.8%)		323 (60.6%)	24 (40.7%)	
> 5	275 (38.2%)	58 (69.0%)		130 (35.7%)	24 (63.2%)		210 (39.4%)	35 (59.3%)	
CA199 (U/ml)			< 0.001			0.003			0.022
0–37	540 (75.0%)	36 (42.9%)		263 (72.3%)	18 (47.4%)		388 (72.8%)	34 (57.6%)	
> 37	180 (25.0%)	48 (57.1%)		101 (27.7%)	20 (52.6%)		145 (27.2%)	25 (42.4%)	
Grade			< 0.001			< 0.001			< 0.001
I-II	524 (72.8%)	29 (34.5%)		253 (69.5%)	13 (34.2%)		392 (73.5%)	21 (35.6%)	
III-IV	196 (27.2%)	55 (65.5%)		111 (30.5%)	25 (65.8%)		141 (26.5%)	38 (64.4%)	
Histology			< 0.001			< 0.001			0.001
Adenocarcinoma	667 (92.6%)	56 (66.7%)		348 (95.6%)	26 (68.4%)		485 (91.0%)	45 (76.3%)	
Mucinous/signet ring-cell	53 (7.36%)	28 (33.3%)		16 (4.40%)	12 (31.6%)		48 (9.01%)	14 (23.7%)	
Tumor size (cm)			< 0.001			0.001			0.010
< 5	424 (58.9%)	29 (34.5%)		209 (57.4%)	11 (28.9%)		275 (51.6%)	20 (33.9%)	
≥ 5	296 (41.1%)	55 (65.5%)		155 (42.6%)	27 (71.1%)		258 (48.4%)	39 (66.1%)	
Age (years)	61.6 (12.1)	61.1 (12.9)	0.765	62.1 (12.3)	59.8 (14.5)	0.363	63.3 (11.8)	60.2 (11.4)	0.050

P = 0.010 for validation set) were significant indicators for OM (Table 2). Subsequently, location, preoperative CEA, preoperative CA199, grade, histology and tumor size were selected for constructing the model.

**Fig. 3** The receiver operator characteristics curves of XGB, ANN, LR, RF, SVM and DT models in (a) Training set, (b) Test set, and (c) Validation set

Model performance

Six ML models including XGB, ANN, RF, LR, SVM and DT were performed in the training set to develop prediction models. In the training set, the XGB model exhibited the best performance (AUC, 0.924, 95%CI, 0.899–0.948, 0.096 gain vs LR) compared to the ANN (AUC, 0.892, 95%CI, 0.858–0.926, $P < 0.001$), RF (AUC, 0.829, 95%CI, 0.777–0.882, P value < 0.001), LR (AUC, 0.828, 95%CI, 0.783–0.873, $P < 0.001$), SVM (AUC,

0.820, 95%CI, 0.769–0.872, P value < 0.001) and DT models (AUC, 0.771, 95%CI, 0.713–0.828, $P < 0.001$) (Fig. 3a). In the test set and the validation set, the AUC results also showed that the XGB model presented the best predictive ability for the OM status (AUC = 0.868 in test set, 0.038 gain vs LR; AUC = 0.766 in validation set, 0.065 gain vs LR) (Fig. 3b-c). The confusion matrix was assessed to show the detailed results of the XGB model and 80%, 76%, and 75% of OM patients were predicted correctly in the training set, test set and validation set,

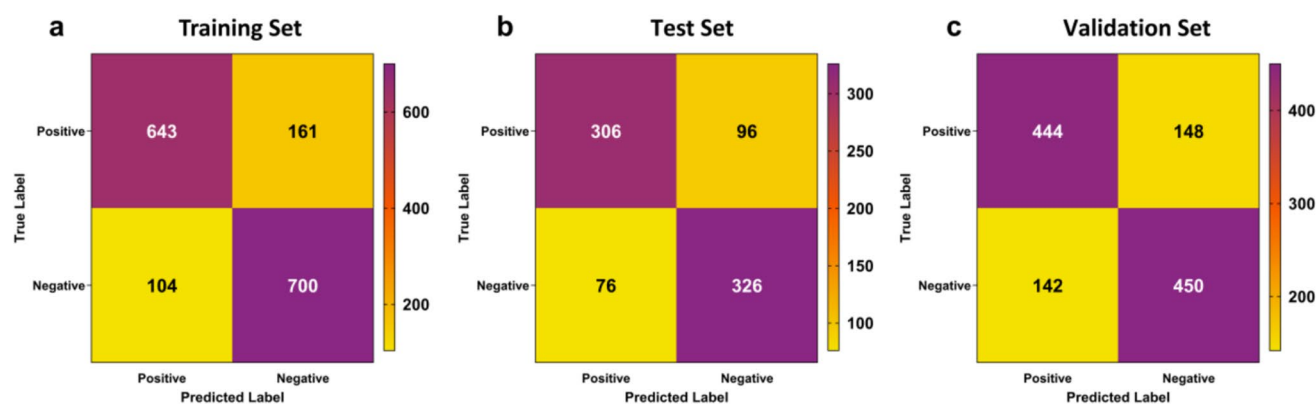


Fig. 4 Confusion matrix of the XGB model in (a) Training set, (b) Test set, and (c) Validation set. Values are normalized across each row

Table 3 Performance of ML models

Model	AUC	Accuracy	Sensitivity	Specificity	Precision	F1 Value
Training Set						
XGB	0.924	0.862	0.798	0.869	0.416	0.547
ANN	0.892	0.835	0.786	0.840	0.365	0.498
RF	0.829	0.803	0.750	0.810	0.315	0.444
LR	0.828	0.775	0.738	0.779	0.281	0.407
SVM	0.820	0.752	0.726	0.756	0.257	0.380
DT	0.771	0.724	0.702	0.726	0.230	0.347
Test Set						
XGB	0.868	0.806	0.763	0.810	0.296	0.426
ANN	0.852	0.774	0.737	0.777	0.257	0.381
RF	0.830	0.744	0.711	0.747	0.227	0.344
LR	0.808	0.692	0.658	0.695	0.184	0.287
SVM	0.763	0.637	0.632	0.637	0.154	0.247
DT	0.739	0.607	0.605	0.607	0.139	0.225
Validation Set						
XGB	0.766	0.762	0.746	0.764	0.259	0.384
ANN	0.752	0.750	0.729	0.752	0.246	0.368
RF	0.706	0.723	0.695	0.726	0.219	0.333
LR	0.701	0.677	0.678	0.677	0.189	0.295
SVM	0.635	0.654	0.661	0.653	0.174	0.276
DT	0.613	0.623	0.627	0.623	0.155	0.249

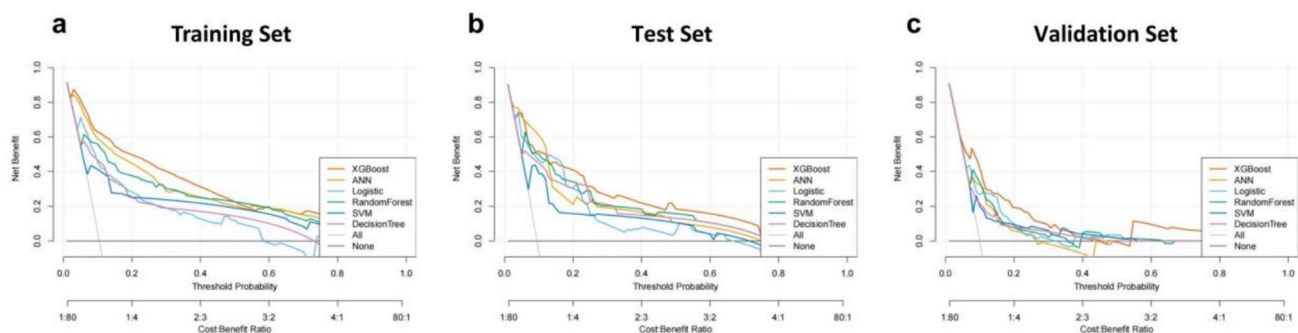


Fig. 5 The decision curve analyses of the XGB, ANN, LR, RF, SVM and DT models in (a) Training set, (b) Test set, and (c) Validation set

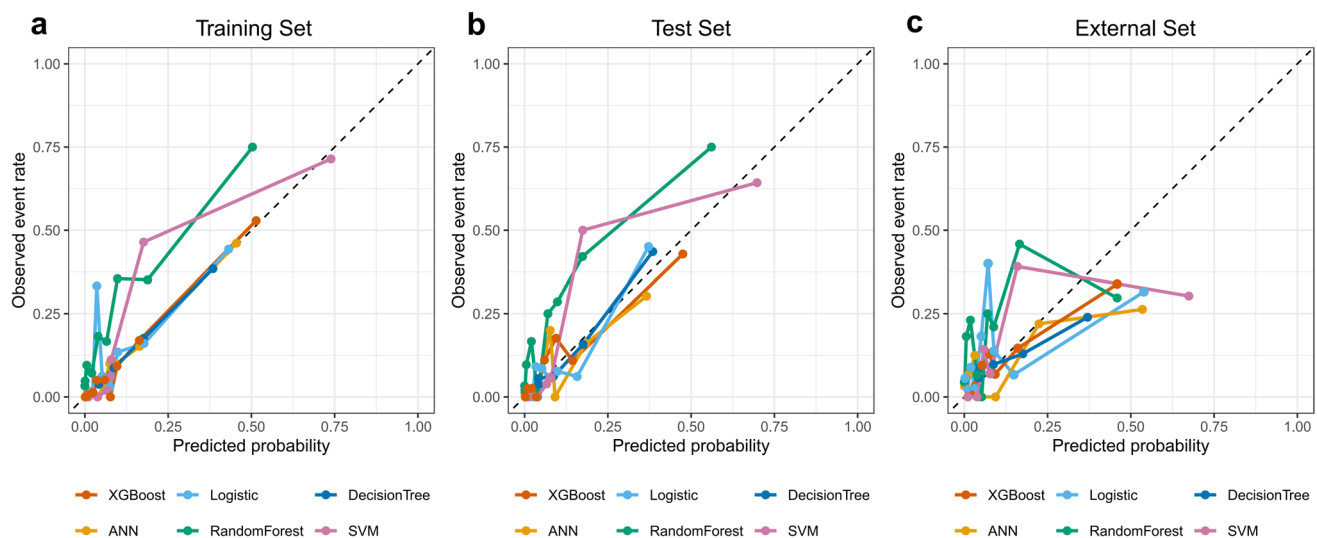


Fig. 6 The calibration plots of the XGB, ANN, LR, RF, SVM and DT models in (a) Training set, (b) Test set, and (c) Validation set

respectively (Fig. 4a-c). Upon detailed analysis of the model performance, the XGB model in training set displayed exceptional performance, achieving an accuracy of 0.862, a sensitivity of 0.798, a specificity of 0.869, a precision value of 0.416 and an F1 score of 0.547, which was also determined in test and validation sets (Table 3). Moreover, Decision curve analysis (DCA) is a decision-theoretic method to evaluate the potential clinical utility of disease diagnostic models. The DCA curve further revealed that the XGB model provided higher net benefit across clinically relevant thresholds of 0.1–0.4 compared other models (Fig. 5a-c). Calibration plots revealed that the XGBoost model exhibited the best performance across the training set, test set, and external validation set, with predicted probabilities closely aligning with observed OM rates (Fig. 6a-c). Conclusively, these findings strongly indicated that the XGB model outperformed the other five models in predicting OM.

Variable importance

Feature importance analysis was assessed to learn about the contribution of each variable to the models. Across different models (XGB, ANN, LR, RF, SVM, DT), tumor location consistently showed relatively high feature importance. It ranked among the top—important features in most models, indicating its significant role across these algorithms for predicting OM in right-sided colon cancer. Other features such as histology, tumor size, Grade, CEA and CA199 had varying importance levels depending on the specific model (Fig. 7).

In the XGB model, tumor location was the most important feature, with a relatively high importance score. Histology and Grade also showed notable importance, while tumor size, CEA, and CA199 have lower but still measurable importance values. Overall, XGB model assigned distinct importance to each of these features, with tumor location standing out as the key one in this model's feature ranking

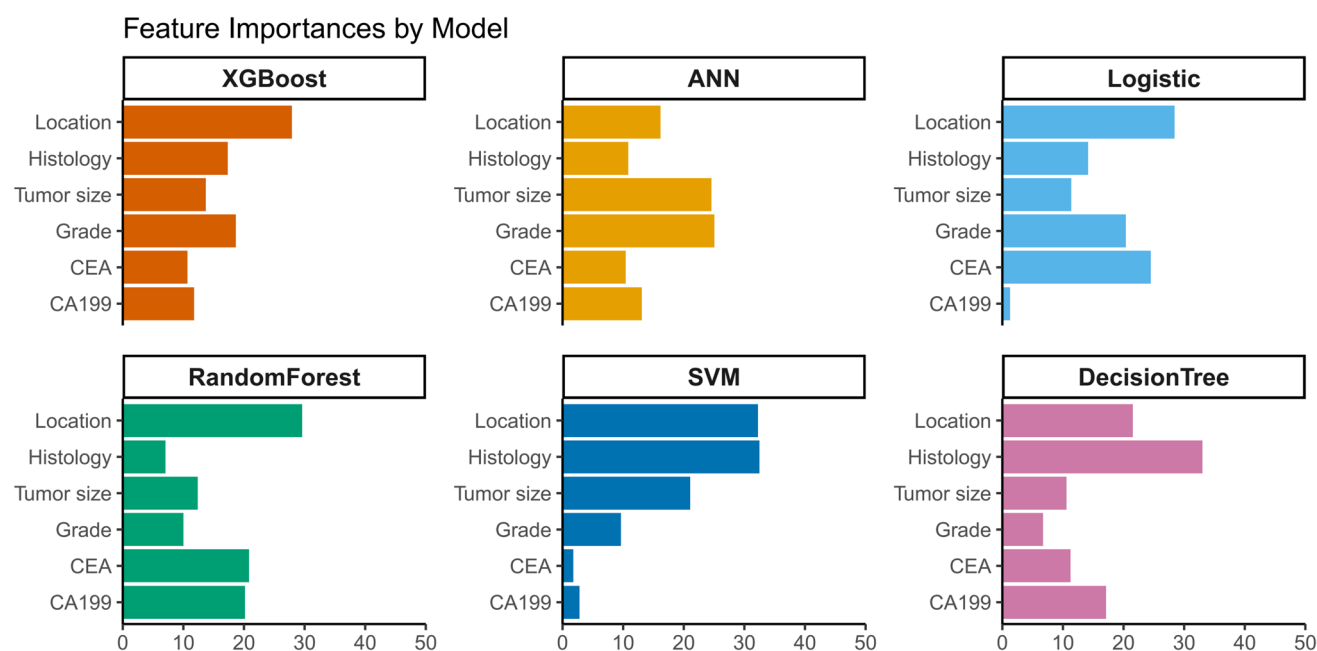


Fig. 7 Feature importance in XGB, ANN, LR, RF, SVM and DT models

(Fig. 7). The SHAP summary plot for the XGBoost model also illustrated the impact of each feature on OM prediction, which also determined that all indicators exerted meaningful effects in the current model (**Fig. S1**).

Discussion

This multicenter retrospective study successfully developed and validated a machine learning (ML)-based predictive model for preoperative identification of omental metastasis (OM) in patients with right-sided colon cancer (RCC). By utilizing routinely available clinical variables (tumor location, preoperative CEA/CA199 levels, grade, histology, and tumor size), we trained six ML algorithms. The XGB model outperforming five other algorithms (ANN, RF, LR, SVM, and DT) across training set (AUC = 0.924), internal test (AUC = 0.868) and validation set (AUC = 0.766). The current model represents a significant advancement in preoperative risk stratification for RCC patients, potentially guiding personalized therapeutic strategies.

Feature importance analysis confirmed that transverse colon cancer was the most significant predictor of OM. Anatomically, the transverse colon is adjacent to the greater omentum, a large peritoneal fold that hangs from the transverse colon and covers abdominal organs, as a result of which, tumors arising here are more likely to invade the serosa and directly seed omental tissue [26, 27]. Other important variables, including histology and tumor grade, consistent with previous studies showing that aggressive

histological subtypes and poorly differentiated tumors are more prone to metastatic diseases [28–30]. Preoperative CEA and CA199, well-established tumor markers, further enhanced predictive performance, reflecting their role in assessing tumor burden and aggressiveness [31]. Together, these variables capture multiple dimensions of RCC biology, highlighting the value of integrating clinical, pathological, and serological factors in OM prediction.

A key distinction of the current study was its exclusive focus on omental metastasis, as opposed to the broader category of peritoneal metastasis. The omentum is a specialized peritoneal structure consisting of a double layer of fatty tissue that hangs from the stomach and covers the abdominal organs, with unique lymphatic and vascular networks that make it a preferential site for metastatic spread in colon cancer [16]. In contrast, peritoneal metastasis refers to tumor dissemination across the entire peritoneal lining, which includes the parietal peritoneum and visceral peritoneum [32]. While omental involvement can coexist with peritoneal metastasis, it represents a distinct clinical entity with specific implications [33]. The omentum's rich vascularity and immune function may accelerate tumor growth and systemic spread, and its anatomical accessibility makes it a target for surgical resection as part of cytoreductive therapy. By focusing specifically on OM, our model addresses a gap in existing literature, which often conflates omental and peritoneal disease. Thus, our model enables more precise risk stratification for this unique metastatic site, guiding tailored interventions that may not be captured by broader peritoneal metastasis prediction tools.

Notably, the clinical significance of this prediction model lies in its potential to guide targeted diagnostic and therapeutic strategies for RCC patients. For patients identified as high-risk for OM by the model, clinicians can implement more intensive preoperative evaluations such as contrast-enhanced CT with thin-slice imaging and PET-CT to better visualize subtle omental lesions that may be missed by conventional imaging [34]. Intraoperatively, high-risk patients can undergo more meticulous omental exploration, including systematic inspection and biopsy of suspicious areas, to confirm or rule out metastasis. This targeted approach ensures that OM is not underdiagnosed, allowing for timely adoption of aggressive treatments like cytoreductive surgery or hyperthermic intraperitoneal chemotherapy when indicated. However, low precision (0.259–0.416) means some high-risk predictions are false positives. Clinicians should combine model outputs with intraoperative exploration to avoid unnecessary omentectomy. It could also integrate with electronic health records (EHRs), automatically extracting routine preoperative variables to generate real-time OM risk scores and concise feature importance summaries. Additionally, linking to clinical decision support systems lets the model offer targeted guidance, such as helping surgeons assess candidates for cytoreductive surgery or HIPEC, tying predictive outputs directly to frontline treatment choices.

This research delivers significant methodological and clinical advancements through its pioneering development of a machine learning-based predictive model for OM in RCC patients. First, it establishes the first dedicated OM prediction tool specifically for RCC patients, addressing a critical gap in existing literature which predominantly focuses on generalized peritoneal metastasis. Second, the multicenter study incorporating geographically distinct cohorts from Zhejiang Cancer Hospital (training/testing) and the Second Affiliated Hospital of Harbin Medical University (external validation), enhancing generalizability and mitigating single-institution bias, with consistent OM prevalence (9.5%–10.4%) and model performance (AUC 0.766–0.924) across cohorts underscoring robustness. Third, the model prioritizes clinical practicality by leveraging six routinely available preoperative variables (tumor location, CEA, CA19-9, grade, histology, tumor size), ensuring immediate clinical application without requiring specialized genomic or radiomic profiling.

The 10% prevalence of omental metastasis (OM) creates notable data imbalance limitations. Since models tend to optimize overall accuracy by default, they prioritize correctly classifying the majority non-OM group over the minority OM group. This leads to suboptimal performance for OM prediction, even the top-performing model shows low precision and modest F1 scores. SMOTE and ensemble calibration may enhance model performance but are not

strictly necessary. SMOTE, which generates synthetic OM samples by linearly interpolating features between minority-class neighbors showed potential for improving precision. Ensemble calibration involves weighting OM samples more heavily during training to reduce bias toward non-OM cases, plus post-hoc calibration to align predicted probabilities with actual OM risk. While these methods could maintain high OM detection sensitivity while lowering false positives, the existing model has already controlled overfitting via 10×tenfold cross-validation and demonstrates basic clinical utility. Thus, SMOTE and ensemble calibration are optional optimizations rather than mandatory fixes.

Limitations of this study should be acknowledged. First, as a retrospective study, it was inevitable to have observer and confusion bias and needed to be verified by some prospective clinical studies. Second, although the validation cohort is geographically distinct, it remains within China. This constrains the demonstrated generalizability of the relevant model, as external validation in non-Chinese or Western populations is required to confirm its broader applicability. Third, potentially relevant variables such as molecular markers which may further improve accuracy. Fourth, over the 18-year period (2006–2024), changes in key clinical elements introduce notable bias. Imaging modalities shifted from standard CT, leading to underdiagnosis in early cases and potential overdiagnosis later. Surgical practices moved from open surgery (limited omental visualization, fewer OM biopsies) to minimally invasive techniques (magnified views, more biopsies), resulting in detection bias as OM confirmation is more likely in later cases. Besides, a key future step for broader generalization, aligned with feedback urging a perspective on multimodal AI (radiomics + clinical data), is integrating the current model's preoperative clinical variables with radiomic features from CT/MRI. This fuses clinical context with imaging's subtle details addressing the current model's single-data-type limitation. Such multimodal AI would better adapt to diverse cohorts and clinical settings, boosting generalization beyond the current study's scope.

In conclusion, our study developed and validated an XGB-based machine learning model that accurately predicts omental metastasis in RCC patients using routine preoperative variables. This model demonstrates strong discriminative performance and potential clinical utility, assisting personalized risk stratification and appropriate treatment decisions. Future prospective multicenter studies with larger cohorts are needed to further validate and refine the model, ultimately improving the management and outcomes of RCC patients at risk of OM.

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Data availability De-identified data and code used for model training are available upon reasonable request to the corresponding author, in accordance with institutional and national data protection regulations.

Declarations

Ethical approval Ethical review board approval was obtained and informed consent was obtained from the institutional review board (2025-602).

Competing interests The authors declare no competing interests.

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